

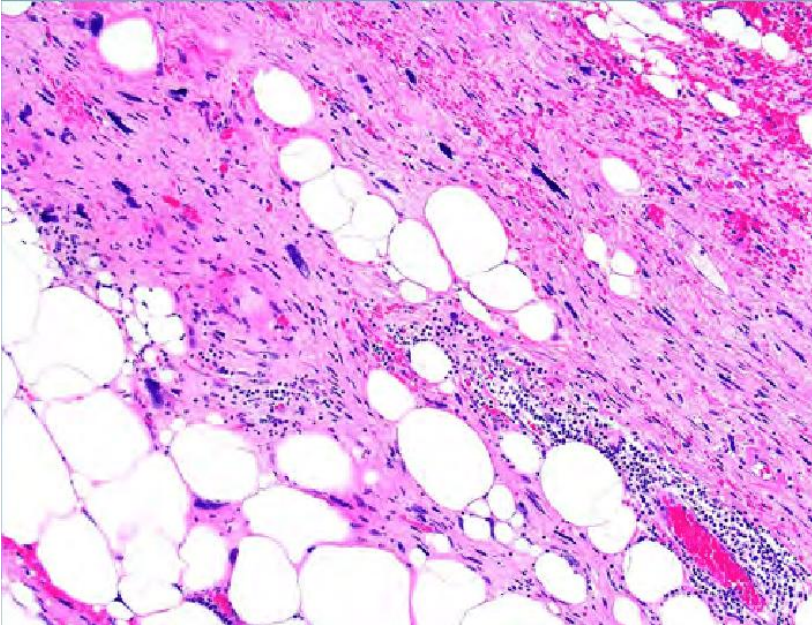
FOUNDATION MODELS FOR CLASSIFICATION OF SOFT TISSUE TUMORS

Carlo Peron



SOFT TISSUE TUMORS

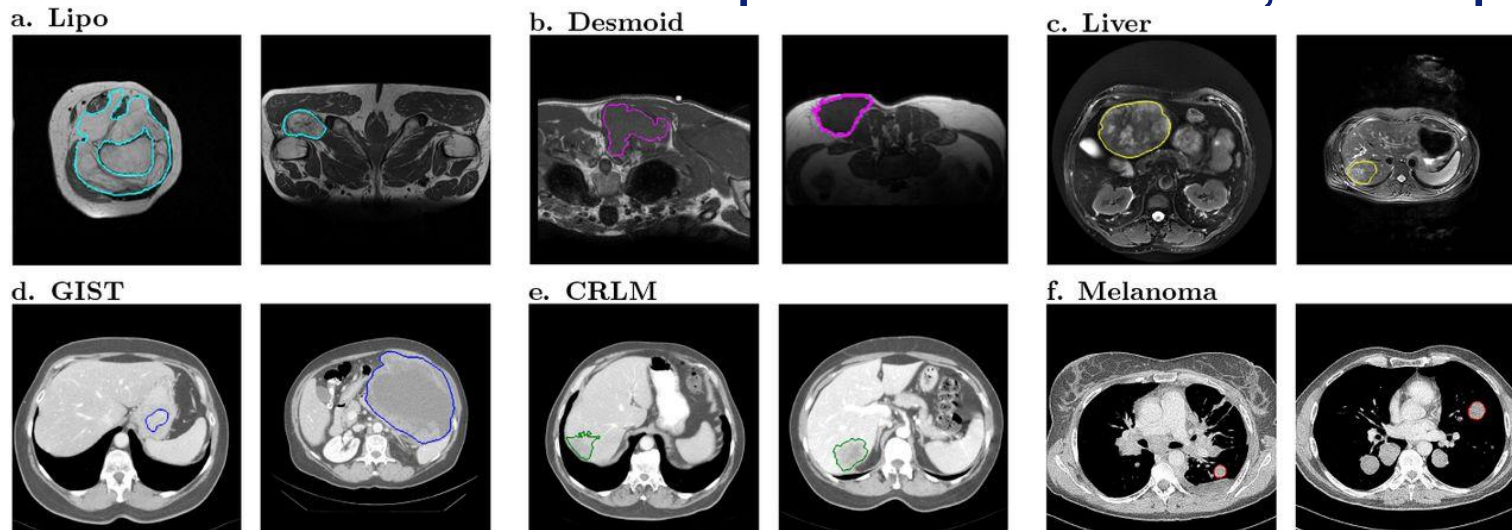
- Heterogeneous group of lesions, the diagnosis often requires invasive biopsy because many tumors look similar on imaging [Fletcher, 2018]
- Key clinical challenge: distinguishing benign from malignant subtypes non-invasively e.g. lipoma vs well-differentiated liposarcoma appears nearly identical on MRI
- Machine learning can help, but faces three barriers: (1) small labeled datasets, (2) high computational cost of 3D volumes, (3) need for generalizable features across scanners and protocols



[Well Differentiated Liposarcoma](#)

WORC

- WORC is a dataset containing 3 soft-tissue tumors classes, that totals 930 patients across 6 tumor categories: GIST (246), Desmoid (203), Liver (186), Lipo (115), Melanoma (103), CRLM (77)
- Each case contains volumetric CT or MRI + expert segmentation mask + clinical label [Starmans et al., 2021]

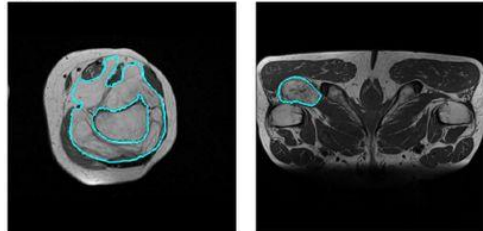


[WORC scans](#)

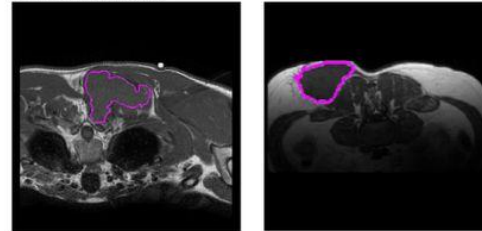
WORC

- Why WORC is hard:
 - Small cohorts (77–246 cases) where deep learning typically needs thousands
 - Subtle imaging differences, tasks are ambiguous even for expert radiologists
 - Mixed modalities (CT and MRI), variable scanners and protocols

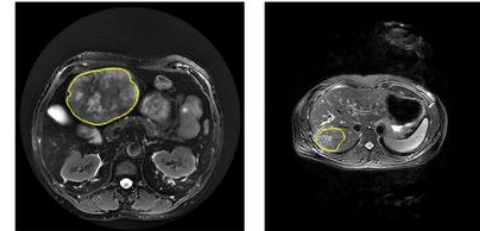
a. Lipo



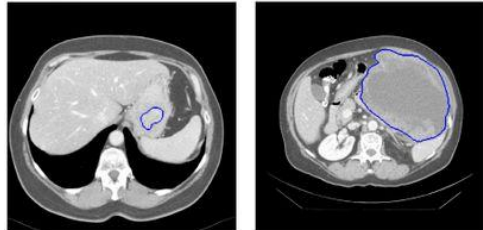
b. Desmoid



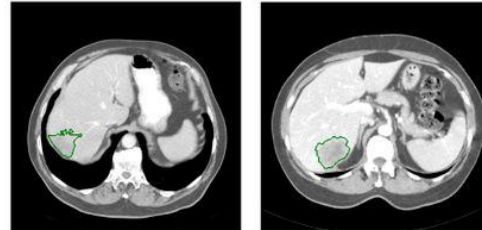
c. Liver



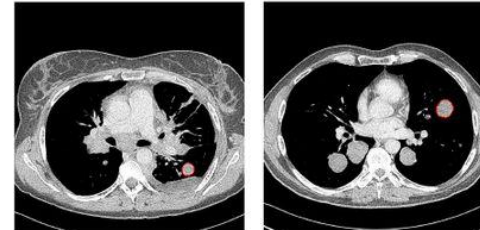
d. GIST



e. CRLM



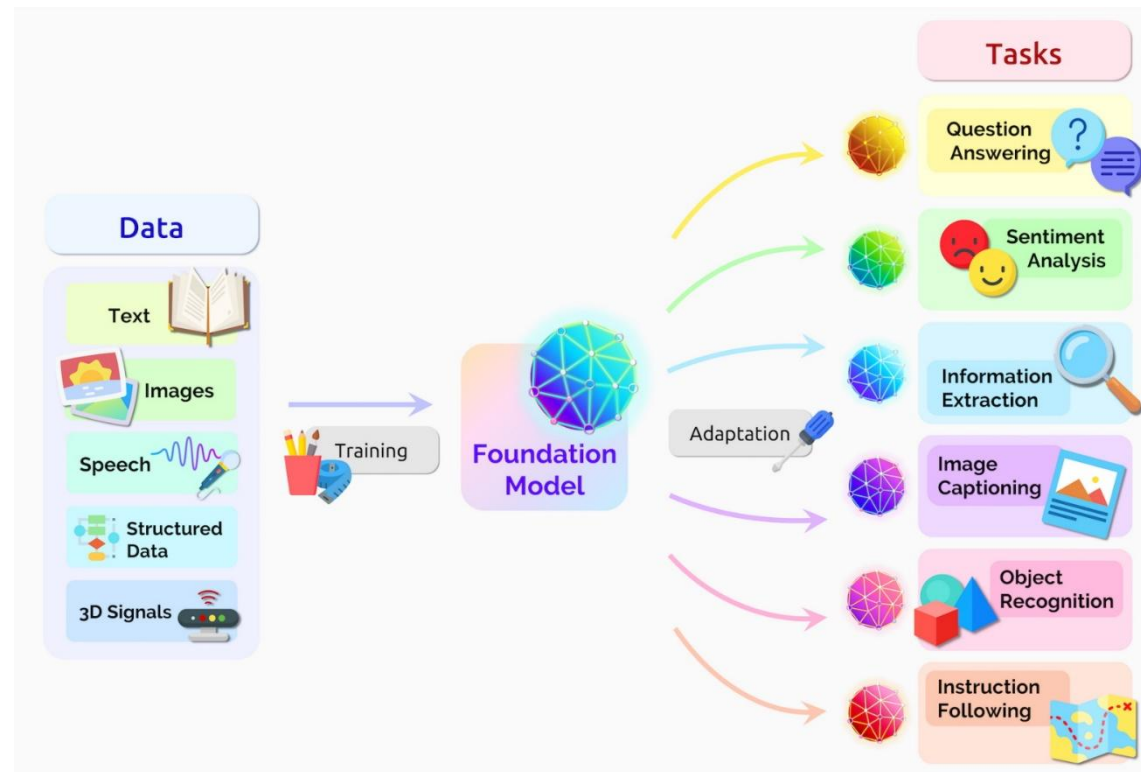
f. Melanoma



[WORC scans](#)

FOUNDATION MODELS

- Foundation models are **pre-trained on massive, diverse datasets** and can be **reused** across downstream tasks with minimal adaptation
- The advantages
 - Time
 - Compute
 - Generalizability
 - Modularity



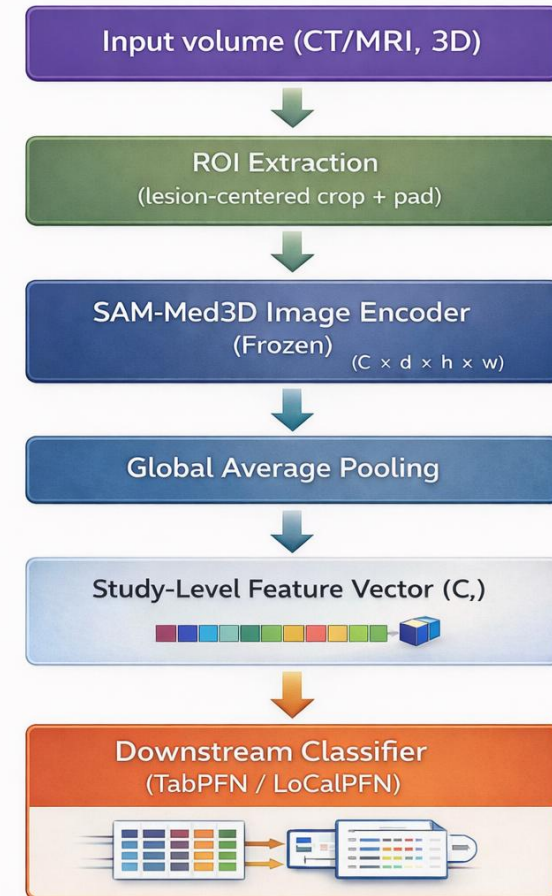
Foundation models

FOUNDATION MODELS

- SAM-Med3D, a segmentation model is used as encoder here requires the input to be $128 \times 128 \times 128$ so there's the need to do some preprocessing
- After having extracted the features we need to vectorize them in order to allow TabPFN, the model used for classification, to work

Feature Extraction Pipeline

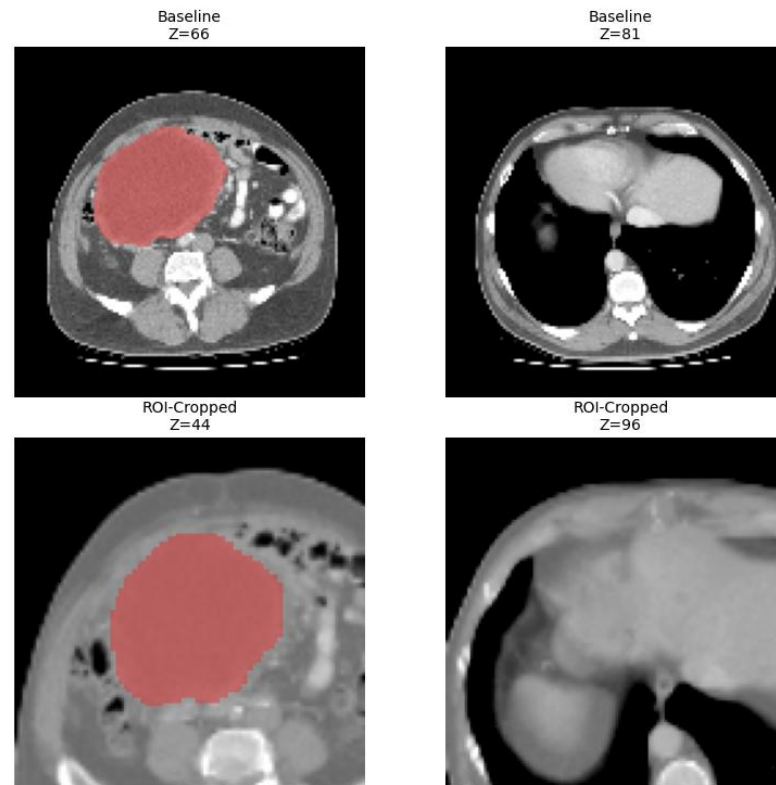
ROI-cropped volumes → SAM-Med3D embeddings → pooled features → tabular classifier



PREPROCESSING

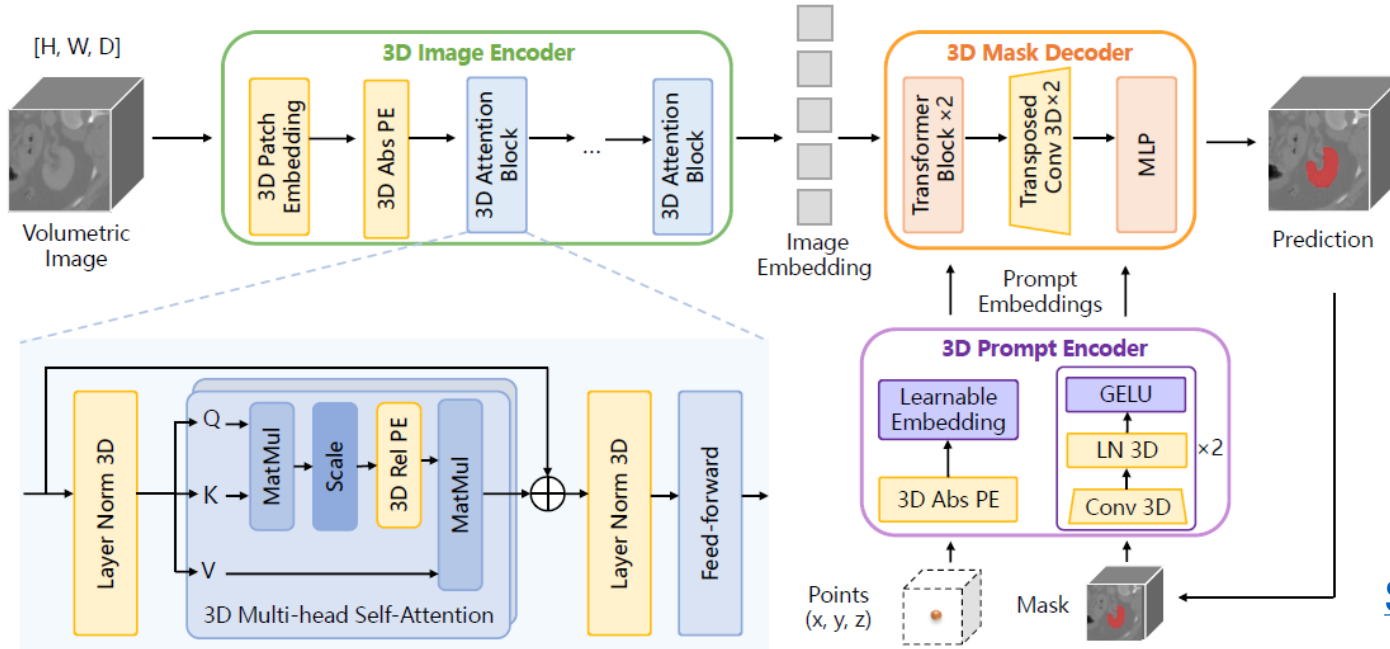
- Three strategies compared all evaluated with TabPFN + percentile pooling + 5-fold CV (Experiment 3):

- Full volume baseline
- Filtered baseline
- ROI cropping



SAM-MED3D

- Foundation model for 3D medical image segmentation it extends Segment Anything Model (SAM) from 2D natural images → 3D medical volumes [Wang et al., 2023]
- It is originally designed for segmentation, but in this study only part of the whole model (the encoder) is used to extract features from the scans



[SAM-Med3D architecture](#)

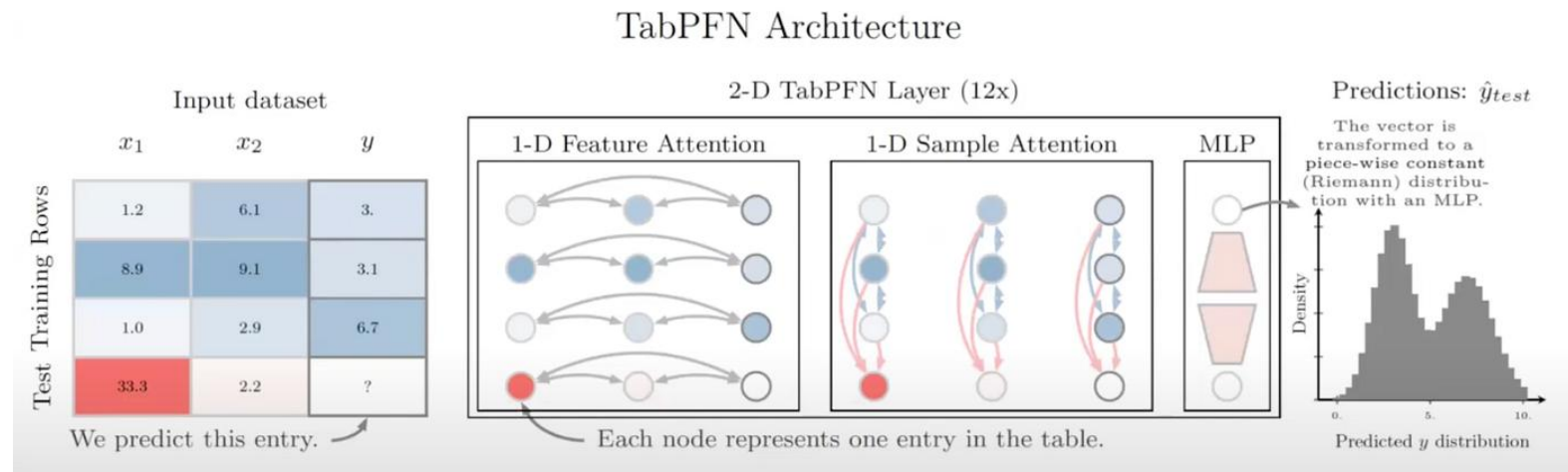
CSDN @liyhao76

POOLING LAYER & STANDARDIZATION

- Encoder output ($384 \times 8 \times 8 \times 8$) must be compressed into a fixed-size vector per case using a pooling layer + PCA
- Three pooling strategies compared, all with TabPFN + ROI cropping + 5-fold CV (Experiment 4):
 - **Average pooling:** global mean $\rightarrow$ 384-dim vector (no PCA needed)
 - **Multiscale pooling:** adaptive average at 1^3 , 2^3 , 4^3 resolutions $\rightarrow$ concatenate $\rightarrow$ 28,032-dim (PCA to ≤ 500)
 - **Percentile pooling:** 10th, 25th, 50th, 75th, 90th percentile per channel $\rightarrow 5 \times 384 = 1,920$ -dim (PCA to ≤ 500)

TABPFN

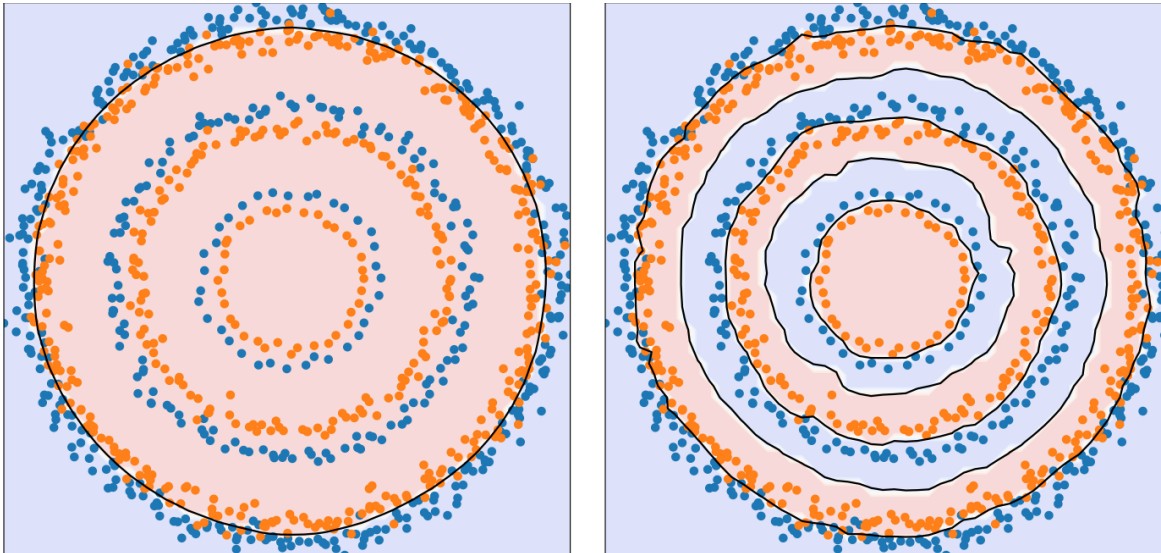
- Tabular Prior-data Fitted Network is a transformer-based classifier for small tabular datasets [Hollmann et al., 2023]
- **In-context learning:** training set + labels + test query concatenated → single forward pass → class probabilities — no gradient updates at inference
- Designed for datasets with <10,000 samples is perfect for WORC cohorts (77–246)



[TabPFN architecture](#)

LOCALPFN

- LoCalPFN = Locally Calibrated PFN — extends TabPFN with local neighborhood retrieval [Thomas et al., 2024]
- Motivation: tumors have subtypes that occupy different regions of feature space with different decision boundaries — a global classifier may miss these [Atkeson et al., 1997]



TabPFN vs LoCalPFN decision boundary

THE EXPERIMENTS

- Experiment 1: Which preprocessing technique is optimal
- Experiment 2: Which pooling layer leads to the best performance
- Experiment 3: Custom methods against baselines
- Experiment 4: Attention maps, does SAM-Med3D look at the lesions?
- Experiment 5: Embeddings diagnostics, do the embeddings carry label-discriminative features?
- Experiment 6: Can a classification head positioned on top of SAM-Med3D outperform TabPFN?

[TabPFN vs LoCalPFN decision boundary](#)

PREPROCESSING

- Full volume baseline, Filtered baseline and ROI cropping were compared to establish which one guaranteed the greatest gains
- ROI cropping with 30-voxel margin appears to be the best performing technique and it is therefore adopted as default for all subsequent experiments

Dataset	Accuracy			Macro F1			ROC AUC		
	Base	Filt	ROI	Base	Filt	ROI	Base	Filt	ROI
CRLM	0.481	—	0.508	0.437	—	0.464	0.454	—	0.491
Desmoid	0.680	0.740	0.694	0.502	0.425	0.613	0.645	0.515	0.659
GIST	0.553	0.574	0.617	0.537	0.381	0.617	0.596	0.563	0.654
Lipo	0.504	0.470	0.504	0.497	0.388	0.497	0.481	0.337	0.466
Liver	0.645	0.667	0.597	0.644	0.400	0.592	0.679	0.558	0.605
Melanoma	0.397	0.533	0.476	0.384	0.467	0.462	0.390	0.300	0.526
Mean	0.543	0.597	0.566	0.500	0.412	0.541	0.541	0.455	0.567

POOLING LAYER & STANDARDIZATION

- Multiscale and percentile comparable in accuracy and F1, but percentile achieved highest mean AUC (0.637 vs 0.624)
- Percentile uses 15× fewer raw features than multiscale → more efficient PCA

Dataset	Accuracy			Macro F1			ROC AUC		
	Avg	Multi	Perc	Avg	Multi	Perc	Avg	Multi	Perc
CRLM	0.376	0.651	0.494	0.349	0.617	0.480	0.406	0.641	0.492
Desmoid	0.656	0.714	0.670	0.537	0.639	0.559	0.670	0.693	0.707
GIST	0.557	0.618	0.642	0.555	0.616	0.641	0.595	0.669	0.710
Lipo	0.591	0.565	0.617	0.586	0.557	0.604	0.604	0.608	0.645
Liver	0.581	0.554	0.586	0.570	0.543	0.582	0.577	0.574	0.619
Melanoma	0.610	0.525	0.611	0.600	0.519	0.603	0.655	0.558	0.650
Mean	0.562	0.604	0.603	0.533	0.582	0.578	0.584	0.624	0.637

TRANSFER LEARNING EXPERIMENT

- The 4 methods compared on all 6 WORC datasets, Med3-TabPFN and Med3-LoCalPFN: frozen SAM-Med3D → percentile pooling → PCA → tabular classifier (5-fold stratified CV)
- 3D DenseNet-121 and 3D ViT: trained from scratch, 50 epochs, cross-entropy loss (single 80/20 split)
- 3D DenseNet-121 dominates: mean Acc = 76.4%, mean AUC = 0.701 showing consistent results across all datasets (69.6%–81.6%)

TRANSFER LEARNING EXPERIMENT

- LoCalPFN failed on the smaller datasets as the formula for calculating the neighborhood is effective only when the dataset size is equal or bigger than 125

Method	Metric	Dataset (<i>n</i>)						Avg
		CRLM (77)	Desmoid (203)	GIST (246)	Lipo (115)	Liver (186)	Melanoma (103)	
Med3-TabPFN	Acc	.467	.650	.585	.496	.527	.551	.546
	AUC	.395	.661	.612	.469	.578	.616	.555
Med3-LoCalPFN	Acc	†	.665	.593	†	.543	†	.600*
	AUC	†	.656	.639	†	.580	†	.625*
3D DenseNet-121	Acc	.750	.780	.780	.696	.816	.762	.764
	AUC	.633	<u>.724</u>	.760	<u>.558</u>	<u>.777</u>	<u>.755</u>	<u>.701</u>
3D ViT	Acc	.688	.756	.700	.348	.632	.333	.576
	AUC	<u>.733</u>	.624	<u>.769</u>	.513	.670	.429	.623

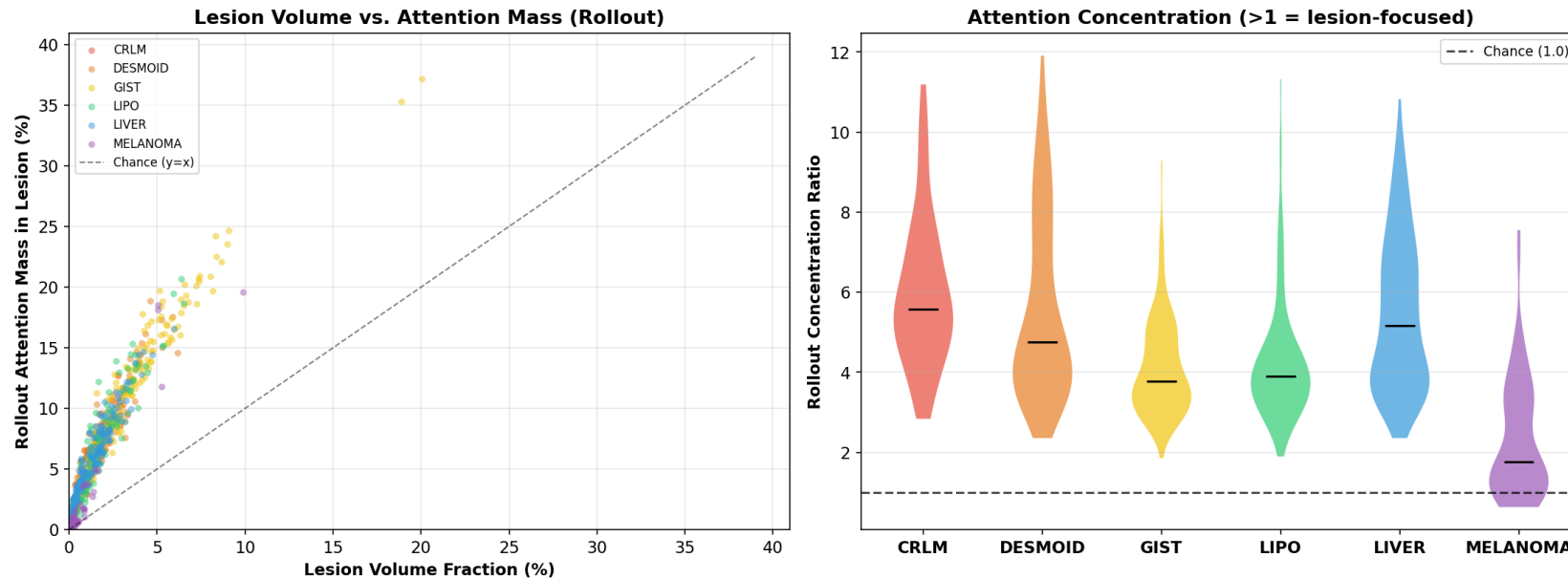
ATTENTION MAPS

- The goal is to determine whether the frozen encoder is even "looking at" the tumor without even taking into consideration the classifier
- This has been achieved by capturing the self-attention weights from all 12 ViT-B blocks using monkey-patching and computing multiplicative rollout across the 4 global attention blocks (2, 5, 8, 11) with residual mixing
- The metrics collected are attention mass (fraction of attention inside lesion mask) and concentration ratio (attention mass $\div$ lesion volume fraction; >1 = above chance)

ATTENTION MAPS

- Mean concentration ratio = $4.1\times$ the encoder consistently attends to lesion regions
- The encoder knows where the tumor is attending to the lesion is necessary, but not sufficient; it must also encode label-discriminative features

SAM-Med3D — Lesion Size vs. Attention Allocation (930 Cases)



EMBEDDINGS DIAGNOSTICS

- **Mann–Whitney U:** tests whether different cases have different values in each of the 1,920 feature dimensions independently
- **HSIC / CKA:** measures whether the entire feature matrix and the label vector are statistically dependent as a whole
- **kNN agreement:** checks whether nearby points in feature space tend to share the same label

EMBEDDINGS DIAGNOSTICS

- **Minimal detectable signal across datasets.** Only Desmoid and GIST showed any class separation
- Where signal exists, it is distributed across many weak dimensions rather than forming separable clusters

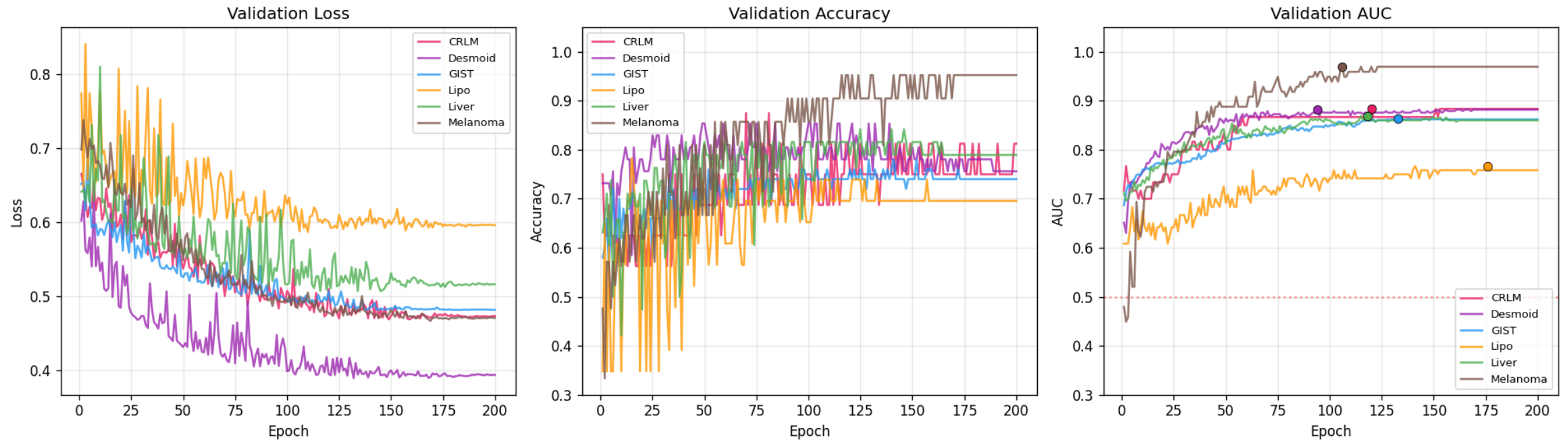
Dataset	n	MW	sig. dims	CKA	HSIC	p	kNN adj. ($k=5$)	kNN p
CRLM	77		0/1920 ●	0.069	0.0830	●	-0.046	0.6793 ●
Desmoid	203		395/1920 ●	0.052	0.0012	●	-0.037	0.7443 ●
GIST	246		679/1920 ●	0.080	0.0001	●	-0.008	0.5255 ●
Lipo	115		0/1920 ●	0.027	0.3245	●	0.113	0.0210 ●
Liver	186		0/1920 ●	0.023	0.1575	●	-0.082	0.9860 ●
Melanoma	103		0/1920 ●	0.033	0.3305	●	-0.037	0.6683 ●

CLASSIFICATION HEAD EXPERIMENT

- Is the bottleneck the encoder or the downstream pipeline?
- Lightweight FC head attached to frozen SAM-Med3D encoder output; only the head is trained (200 epochs, AdamW, $\text{lr} = 10^{-3}$, dropout = 0.3); same ROI-cropped 128^3 inputs
- The embedding diagnostics said "no signal" but the classification head found strong signal in the same features. The encoder distributes class-relevant information across many dimensions in a collectively exploitable but individually weak pattern

CLASSIFICATION HEAD EXPERIMENT

Classification Head — All Datasets (Frozen SAM-Med3D, ROI, 200 epochs)



CLASSIFICATION HEAD EXPERIMENT

- Mean AUC = **0.872**, mean Acc = **79.1%**
- CRLM: AUC jumped from 0.395 (TabPFN) to 0.883 — largest recovery (+0.488)

Dataset (n)	Best Ep.	Acc	AUC	F_1^B	F_1^M	Macro F_1
CRLM (77)	120	.813	.883	.84	.77	.81
Desmoid (203)	94	.756	.882	.83	.55	.69
GIST (246)	133	.740	.864	.71	.76	.74
Lipo (115)	176	.696	.767	.53	.77	.65
Liver (186)	118	.789	.869	.83	.71	.77
Melanoma (103)	106	.952	.969	.96	.93	.95
Mean	124	.791	.872	.78	.75	.77

CONCLUSION

- Transfer-learning pipeline (Exp 1): mean AUC = 0.555
- Classification head on same frozen encoder (Exp 2): mean AUC = 0.872
- Bottleneck is downstream: PCA → tabular classifier
- The encoder distributes class cues across many weak dimensions